

Project Proposal

Predicting Term-Deposit Subscription from Bank Telemarketing Data

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GitHub repository: <https://github.com/Ghaniaali/ML-Preprocessing>

1. Problem Statement

A Portuguese bank sells term deposits through phone campaigns. Only about 12% of contacted clients subscribe, yet every call costs staff time and money. We will build a model that estimates, **before a call is made**, how likely a client is to subscribe, so that the bank can prioritise likely subscribers and reduce wasted calls.

2. Dataset

UCI Bank Marketing dataset (*bank-full.csv*; Moro, Cortez & Rita, 2014): 45,211 client records and 16 input features. Numeric: age, balance, day, campaign, pdays, previous. Categorical: job, marital, education, contact, month, poutcome. Binary: default, housing, loan. The data is public, released under CC BY 4.0, and contains no direct personal identifiers. It is well above the 1,000-row minimum and mixes numeric and categorical features.

3. Target Variable and Task

Target: y (client subscribed to a term deposit: yes = 1, no = 0; 11.7% positive). **Task:** binary classification.

4. Success Metric

Primary metrics are **ROC-AUC** and **PR-AUC** (quality of the ranking of clients), with **F1 on the positive class** as the decision-level measure. Accuracy is not used: a model that always predicts “no” would score about 88% while finding no subscribers.

5. Part 1 Approach (Leakage-Free Pipeline)

The data is split 80/20 with stratification before any learned preprocessing. A scikit-learn *Pipeline* / *ColumnTransformer* then applies median, constant and most-frequent imputation, 1st–99th percentile clipping of skewed features (balance, campaign, previous), standardisation, one-hot encoding, class weighting and SelectKBest feature selection, all fitted on training data only. Baseline 5-fold cross-validation on the training set gives ROC-AUC ≈ 0.75 and PR-AUC ≈ 0.38 (a no-skill model scores about 0.12).

6. Known Risks and How We Handle Them

- **Target leakage:** call duration is only known after the call, so *duration* is removed.
- **Class imbalance:** addressed with class weighting and imbalance-aware metrics.
- **Missing values:** “unknown” entries (poutcome $\approx 82\%$, contact $\approx 29\%$) are meaningful and kept as their own category; rare gaps in job and education are imputed.
- **Zero-inflated features:** pdays and previous are 81.7% constant, so a previously-contacted flag is added and percentile clipping replaces the IQR rule.
- **Evaluation realism:** rows are ordered roughly by time while we use a random split; we note this as a limitation.

7. Suitability for Later Parts

The dataset’s size supports Naive Bayes and decision-tree baselines (Part 2), ensemble and gradient-boosting tuning (Part 3), a multilayer perceptron, and PCA or clustering to segment clients (Part 4). Features such as age and marital status also allow a meaningful fairness and interpretability discussion.